

Trail Running — Training and Race Preparation Guide

What Sets Trail Running Apart

Unlike road running, trail running takes place on unpaved terrain such as forest paths, mountain trails, and technical singletrack, often with significant elevation gain. Races are commonly categorized by distance: short trail races are under 25 km, while ultramarathons typically start at 50 km and can extend beyond 160 km in a single stage.

Elevation Training

A common metric used in trail race planning is elevation gain per kilometer. Technical mountain races can involve over 50 meters of gain per kilometer, compared to under 5 meters per kilometer for flatter courses. Runners preparing for high-elevation-gain races are advised to include at least 2 dedicated hill or stair workouts per week for 8-12 weeks before race day.

Footwear Considerations

Trail running shoes differ from road shoes primarily in outsole lug depth, which is typically 3-5mm for mixed terrain and can exceed 6mm for muddy or technical trails. Most trail shoes also include a rock plate, a semi-rigid layer that protects the foot from sharp rocks and roots underfoot.

Nutrition During Long Races

For races lasting longer than 90 minutes, sports nutrition guidelines commonly recommend consuming 30-60 grams of carbohydrate per hour to maintain energy levels. For ultramarathons exceeding 6 hours, this can increase to 60-90 grams per hour, often split between gels, chews, and real food available at aid stations.

Hydration Guidelines

A commonly cited hydration target for endurance runners is 400-800 ml of fluid per hour, adjusted for heat, humidity, and individual sweat rate. Runners are also advised to replace electrolytes, particularly sodium, during races longer than 2-3 hours to reduce the risk of hyponatremia.

Mandatory Gear in Ultramarathons

Many ultramarathon races, particularly those in mountainous terrain, require runners to carry mandatory gear including a minimum 1-litre water capacity, a emergency space blanket, a whistle, and a fully charged mobile phone. Failure to present mandatory gear at a checkpoint can result in time penalties or disqualification.